MICROECONOMICS · PROSPERITY AND INEQUALITY

Technological Change, Population & Growth

Unit 2 — Study Guide & Lecture Notes

- Decision-making, opportunity cost & economic rent
- Specialization & comparative advantage
- Technology choice, isocost lines & the Industrial Revolution
- Economic models & the climate challenge

Academic year 2025–2026 · Companion to the lecture slides & CORE handbook

BEFORE YOU BEGIN

How to use these notes

These notes follow the lecture slides one-to-one, but expand the reasoning so you can read them on their own, annotate them during class, and revise from them before the exam. Every important graph from the slides has been redrawn cleanly so you can see exactly what the lecturer is pointing at.

DEFINITION BOX

The precise definition you are expected to reproduce in the exam.

WORKED EXAMPLE

Numbers carried all the way through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points the lecturer usually stresses verbally — leave room to add your own notes here.

CHECK YOURSELF

Quick questions to test recall before moving on.

The big question of this unit

For almost all of human history, living standards barely moved. Then, roughly **200–250 years ago**, income per person in a handful of countries began to rise at a rate never seen before — the famous **"hockey-stick"** of growth. This unit builds the economic toolkit needed to answer two linked questions:

1. Why did living standards and population **stagnate** for centuries (the **Malthusian trap**)?
2. Why did they suddenly **take off** with the Industrial Revolution — and why in **Britain** first?

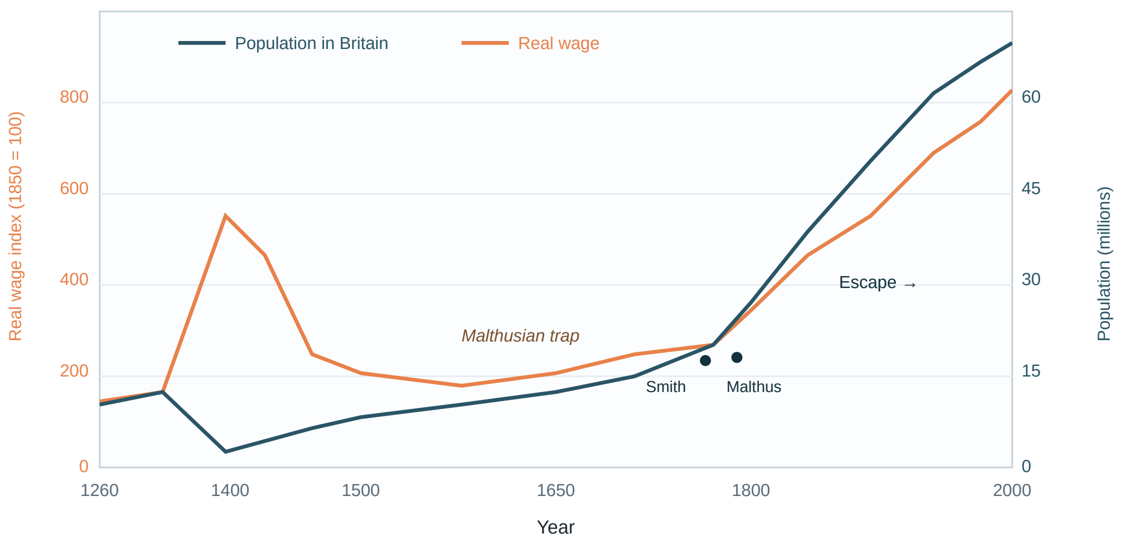

Figure 1. The escape from stagnation. For ~500 years real wages and population were locked in the **Malthusian trap** (flat, with wages falling whenever population rose). From ~1800 both wages *and* population climb together — the Industrial Revolution breaks the trap.

LISTEN FOR IN THE LECTURE

The lecturer uses this chart to frame the whole unit: *autarky vs. trade, Malthus vs. Smith, and why "escape" happens exactly when technology starts improving permanently.* Note the brief dip around 1350 — the Black Death killed a third of the population, so the survivors' real wages jumped. That single event is the classic illustration of the Malthusian mechanism.

SECTION A

Introduction

The rapid, sustained rise in income and living standards we take for granted today is, historically speaking, **brand new**. It is largely the product of **technological progress** — our growing ability to produce more output from the same inputs.

Two puzzles the unit will solve

- **How did the technological revolution start?** What changed so that inventions began to be adopted, one after another, in a self-sustaining wave?
- **Why did it not start earlier?** Humans were clever for millennia — water wheels, windmills and gunpowder all pre-date 1750. So why did take-off wait until the 18th century?

The slide's three images — a woman spinning by hand, the **spinning jenny**, and a mechanised **cotton mill** — trace exactly this transition: from home production, to a single clever machine, to the factory system. Textiles were the first industry to be transformed, and it is the industry we will keep returning to when we model technology choice in Section D.

KEY IDEA OF THE UNIT

We will use **economic models** to explain (i) the centuries of **stagnation** before 1800 and (ii) the **explosive growth** in living standards and population since. The tools introduced along the way — opportunity cost, comparative advantage, production functions and isocost lines — are the core of microeconomics.

LISTEN FOR IN THE LECTURE

The framing question "*Why an island off the coast of Europe, and why then?*" recurs throughout. Keep it in mind: by the end of Section D you should be able to answer it with a diagram, not just a story.

SECTION B

Decision Making

Before we can explain the behaviour of firms and inventors, we need a clean theory of **how a rational agent chooses**. Economics models every decision as a **trade-off between the benefits and the costs** of an action. The subtlety — and the whole point of this section — is that "cost" means more than the money that leaves your wallet.

Direct costs vs. opportunity costs

DEFINITION — OPPORTUNITY COST

The **opportunity cost** of an action is the **net benefit of the next-best alternative action** that you give up in order to take it. It is a cost even though no money changes hands.

- Part of the opportunity cost of **studying** is the **wage you could have earned** working instead.
- The opportunity cost of **starting your own business** is the **salary you could earn as an employee**.

DEFINITION — ECONOMIC COST

$$\textit{Economic cost} = \textit{Direct (out-of-pocket) cost} + \textit{Opportunity cost}$$

A decision is worth taking only if its benefit exceeds this *full* economic cost — not merely the accounting cost.

Worked example — the clothing shop in Brussels

A woman owns a clothing shop on the *Nieuwstraat*. The facts:

- Expected **revenue** this year: €400,000
- **Material expenses**: €300,000
- The **building** is worth €1,000,000. She inherited it 20 years ago and its value is fully written off (so accounting shows no rent cost).
- She could invest money in a fund earning **5% per year**.
- She could earn **€60,000 per year** working elsewhere.

Income Statement (Accounting)	
Revenue	400,000
Material expenses	− 300,000
Accounting profit	**100,000**

Economic Profit	
Revenue	400,000
Material expenses	− 300,000
Accounting profit	**100,000**
Opportunity costs	
Building (5% × €1m)	− 50,000
Forgone wage	− 60,000
Economic profit	**− 10,000**

The **accounting profit is a healthy €100,000**. But the best alternative course of action — **sell the building, invest the €1m, and take a job** — would yield €50,000 (interest) + €60,000 (wage) = **€110,000**. Because that alternative earns more, keeping the shop open actually **destroys €10,000 of value per year**. The **economic profit (or economic rent)** of staying is **−€10,000**.

> WHY THE BUILDING MATTERS EVEN THOUGH IT'S "FREE"
>
> Owning the building outright does *not* make it costless. Every year she holds it, she gives up the €50,000 it could earn if sold and invested. This forgone return is a genuine opportunity cost — a favourite exam trap.

The decision rule

Two equivalent ways to state the same rule:

1. Take the action if **benefit > economic cost** (direct cost + opportunity cost).
2. Take the action if its **net benefit** (benefit − direct cost) exceeds the **net benefit of the next-best alternative** (the opportunity cost).

In the example: net benefit of the shop = **€100,000**; net benefit of the alternative = **€110,000**. Since 100,000 < 110,000, she should **not** keep the shop.

Economic rent and innovation rent

DEFINITION — ECONOMIC RENT

$$\textit{Economic rent} = \textit{benefit} - \textit{direct costs} - \textit{opportunity costs} = \textit{net benefit} - \textit{opportunity cost}$$

A **positive economic rent** means the action beats every alternative; it is the reward for doing something better than the next-best option.

DEFINITION — INNOVATION RENT

A special case of economic rent. The **innovation rent** is the difference between the profit earned with a **new technology** and the profit you would have earned **sticking with the old one**. It is the carrot that gives firms an **incentive** to innovate — and it is the engine that will drive the whole Industrial-Revolution model in Section D.

CHECK YOURSELF

- If the fund's return rose to 8%, what would the shop's economic profit become? *(Answer: opportunity cost of the building = €80,000, so economic profit = 100,000 − 80,000 − 60,000 = −€40,000 — even worse.)*
- Why can a firm show an accounting profit yet still be "failing" economically?

SECTION C

Specialization & Comparative Advantage

Technological progress and rising output are inseparable from **specialization** — people and firms concentrating on a narrow range of tasks and trading for everything else. This section explains *why* specialization raises output, and the surprising result that trade benefits **everyone**, even a producer who is worse at everything.

The gains from specialization

The rise of **firms** lets people specialize in what they are good at. Two mechanisms:

- **Division of labour within firms** — a firm splits one product into many small tasks.
- **Firms focusing on what they do best** — specialization *between* firms.

Specialization raises the **productivity of labour** for three reasons:

THREE SOURCES OF THE GAINS

- **Learning by doing** — repetition makes you faster and better at a task.
- **Differences in ability** — people are naturally better at different things.
- **Economies of scale** — producing many units is more cost-effective per unit than producing few.

Crucially, people can **only** specialize if they have a way to acquire everything else they need. In a capitalist society that mechanism is the **market**.

Adam Smith's pin factory

Adam Smith (*The Wealth of Nations*, 1776) gave the classic illustration. Making a pin involves **18 distinct operations** (drawing the wire, straightening, cutting, pointing, grinding…). Instead of one worker doing all 18:

- A single worker doing everything might make **~20 pins/day**.
- **10 workers**, each specialising in a few operations, produce **48,000 pins/day** — thousands of times more per person.

Why? Time is saved by not switching tasks, skill rises through **learning by doing**, and each person does the operation they are best at. And it does not stop there — the same logic drives **further mechanisation**:

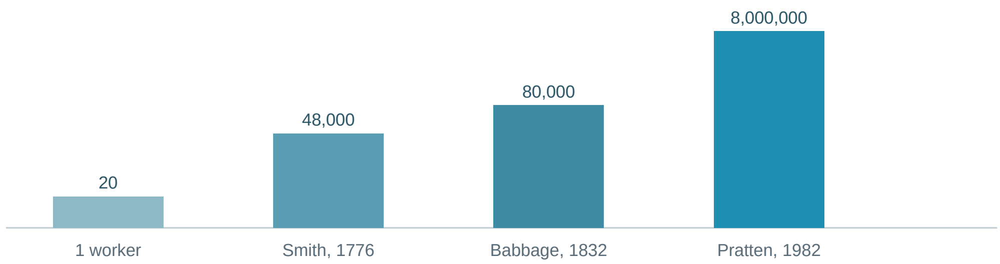

Figure 2. Output per pin-factory day, on a log scale. Specialization (1776) plus two centuries of mechanisation multiplied productivity by a factor of hundreds of thousands. Technology and the division of labour reinforce each other.

Absolute vs. comparative advantage

Now the deep result. When abilities differ, **all** producers gain by specialising — *even if one of them could produce every good more cheaply*. What matters is not who is better in absolute terms, but who has the lower **relative** (opportunity) cost. David Ricardo (1817) proved this with the famous Portugal–England example.

Hours of labour needed	Wine (per unit)	Clothes (per unit)
England	120	100
Portugal	80	90

These are **labour requirements** (the inverse of labour productivity). Portugal needs fewer hours for *both* goods, so Portugal has an **absolute advantage** in everything. It looks like Portugal has nothing to gain from England. Ricardo showed the opposite.

DEFINITION — COMPARATIVE ADVANTAGE

A country (or person) has a **comparative advantage** in a good if its **opportunity cost** of producing that good — how much of the *other* good it must give up — is **lower** than the other country's. A country can be worse at producing both goods (absolute disadvantage) yet still hold a comparative advantage in one of them.

Production Possibilities Frontier (PPF)

Suppose each country has **360,000 working hours**. Devoting them all to one good gives the **maximum production**:

- **England:** 3,000 wine *or* 3,600 clothes.
- **Portugal:** 4,500 wine *or* 4,000 clothes.

All in-between mixes lie on a straight line, the **production possibilities frontier (feasible frontier)**. For England the labour constraint is $100 \cdot \text{Clothes} + 120 \cdot \text{Wine} \le 360{,}000$, i.e.

$$Clothes = 3600 - \frac{6}{5} \cdot Wine \quad (England) \quad Clothes = 4000 - \frac{8}{9} \cdot Wine \quad (Portugal)$$

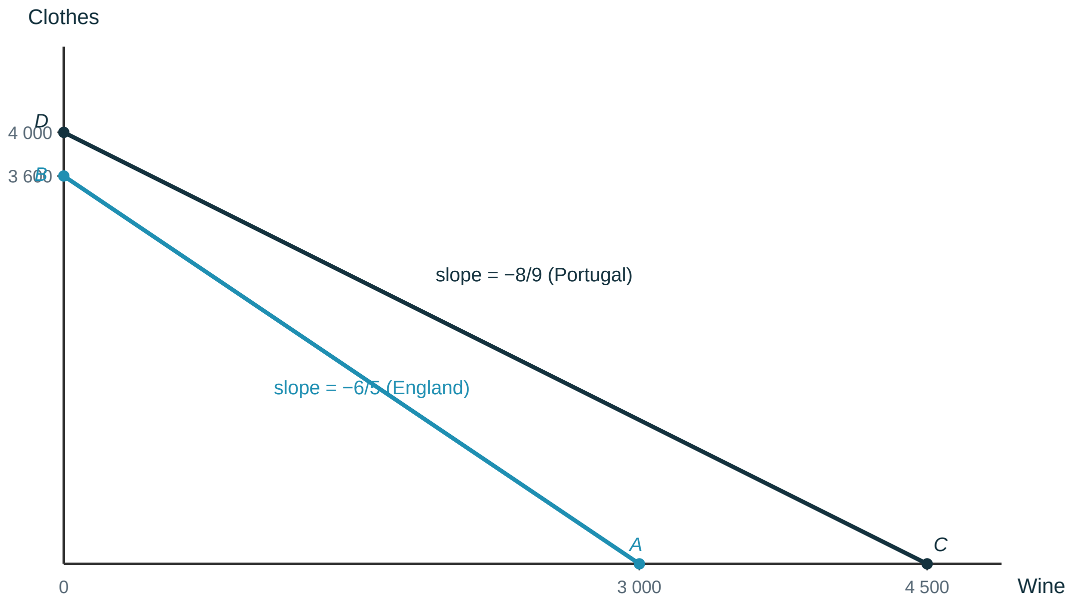

Figure 3. PPFs for the two countries. The **slope of each line is the opportunity cost of wine** (clothes forgone per unit of wine). Portugal's line is flatter (−8/9): giving up wine costs it less clothing than it costs England (−6/5).

Reading off the comparative advantage

	Opportunity cost of 1 wine	Opportunity cost of 1 clothes
England	120/100 = 6/5 clothes	100/120 = 5/6 wine
Portugal	80/90 = 8/9 clothes	90/80 = 9/8 wine

- **Portugal** gives up only **8/9** clothes per wine vs England's **6/5** → **Portugal has the comparative advantage in wine.**
- **England** gives up only **5/6** wine per clothes vs Portugal's **9/8** → **England has the comparative advantage in clothes.**

So the pattern of trade is: **Portugal specializes in wine, England in clothes, and they swap.** Comparative advantages create the *opportunities* for specialization and trade.

The terms of trade and the gains

What price (of wine in terms of clothes) will they trade at? It must sit **between the two opportunity costs**:

$$\frac{8}{9} < price\ of\ wine\ (in\ clothes) < \frac{6}{5}$$

- Below 8/9 → Portugal would rather make its own clothes.
- Above 6/5 → England would rather make its own wine.

Suppose they agree on the simple rate **1 wine = 1 clothes**. Both can now consume **beyond their own PPF**:

> **WORKED EXAMPLE — ENGLAND GAINS FROM TRADE**
>
> Under **autarky**, England might produce and consume 3,000 clothes and 500 wine (labour used = 3000·100 + 500·120 = 360,000 ✓). With trade, England **specialises fully in clothes (3,600)**, keeps 3,000 for itself, and trades **600 clothes for 600 wine** with Portugal. Result: 3,000 clothes **and** 600 wine — more of one good, same of the other. Portugal gains symmetrically.

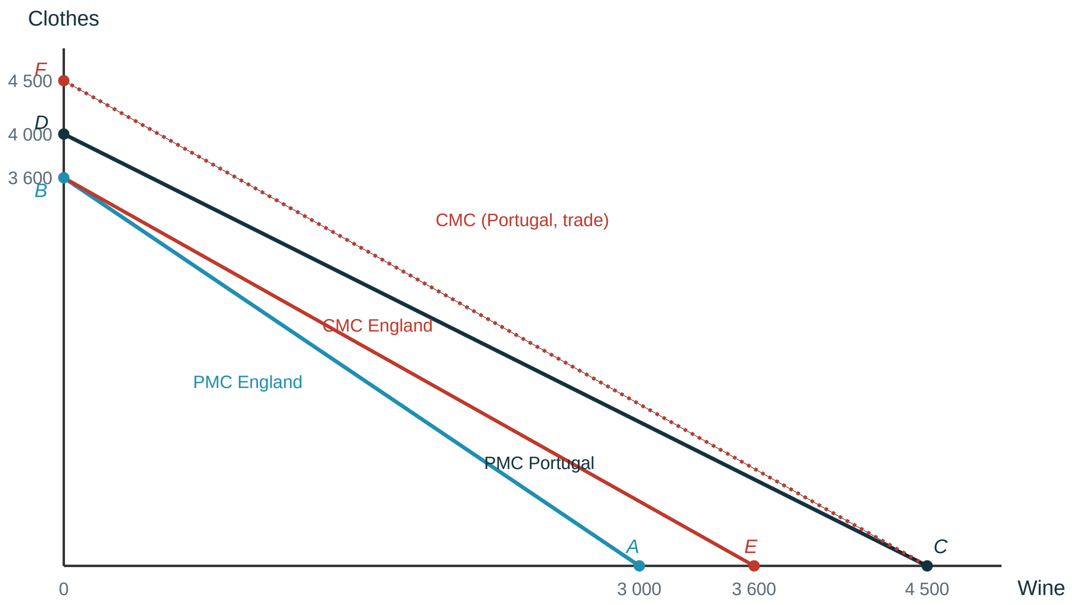

Figure 4. Production (solid) vs **consumption** possibilities (red) once trade opens at 1:1. Each country's red consumption line lies **outside its own production frontier** — the visual proof of the gains from trade. The same logic applies to **two individuals** inside one country, not just to nations.

> **CHECK YOURSELF**
>
> - Portugal has an absolute advantage in both goods — so why does it still gain by trading with England?
> - If the trade price were exactly 6/5 clothes per wine, which country would capture *all* the gains?
> - What does the **slope** of a straight-line PPF represent, and why is it constant here?

SECTION D

Technologies & Innovation

This is the heart of the unit. We build a **model of the firm's technology choice** and use it to answer the framing question: *why did the Industrial Revolution start in 18th-century Britain?* The many possible explanations — high wages, cheap coal, the Enlightenment, culture, colonies — are narrowed to a mechanism we can **draw**: a change in **relative input prices** made energy-hungry machines profitable, and Britain happened to have exactly the right prices.

Firms, technology and the production function

DEFINITIONS

- **Technology** — the method a firm uses to convert inputs into output.
- **Factors of production** — the inputs (labour, machines, energy, land…).
- **Production function** — the relationship between inputs and output, e.g. `Y = f(X)` for one input, or `Y = f(M, N, E)` for machines, workers and energy.

Fixed vs. variable proportions

Consider a simple **olive-oil** technology: three machines (milling, mashing, pressing), one worker to operate them, using 80 kWh, producing 50 litres.

Machines M	Workers N	Energy E (kWh)	Output Y (litres)
3	1	80	50
6	2	160	100
9	3	240	150
12	4	320	200

TWO KEY PROPERTIES

- **Fixed-proportions technology:** to raise output the inputs must be scaled up *together* in fixed ratios (1 worker : 3 machines : 80 kWh). You cannot swap one for another.
- **Constant returns to scale:** doubling every input exactly doubles output.

By contrast, the **food/apples** production of the previous unit was **variable-proportions**: you could add labour to a *fixed* plot of land, giving the familiar **concave** production function (diminishing returns to labour). Fixed proportions is a special case where no substitution is possible.

Graphing a two-input technology

Output depends on two inputs, so a full picture is 3-D and awkward. The trick: put the **two inputs on the axes** and mark a point for each output level. With fixed proportions, equal output levels line up on a **straight ray through the origin**.

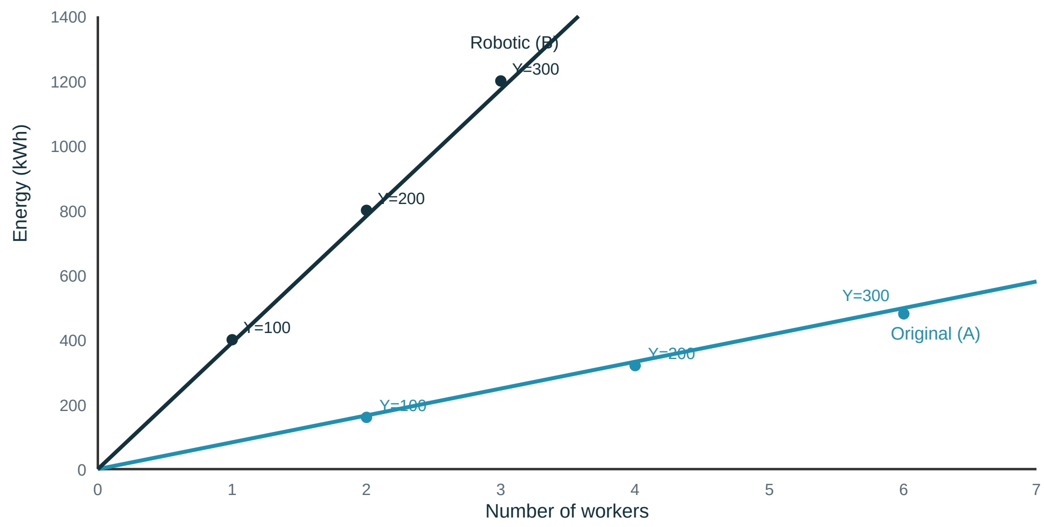

Figure 5. Two **fixed-proportions** technologies for olive oil. The **original (A)** uses more **labour**; the **robotic (B)** uses more **energy** but only 1 worker per 100 litres. Neither is "better" in the abstract — the best choice depends entirely on the **relative prices** of labour and energy.

LISTEN FOR IN THE LECTURE

The point the lecturer is driving at: a technology is just a **bundle of input requirements**. "Which is better?" is **not** answerable until we know input prices. That is the bridge to isocost lines.

Modelling the choice of technology (cloth)

Back to the Industrial Revolution. There are **five ways to produce 100 metres of cloth**, each using labour (workers) and energy (tonnes of coal). Technology **E is labour-intensive**; technology **A is energy-intensive**.

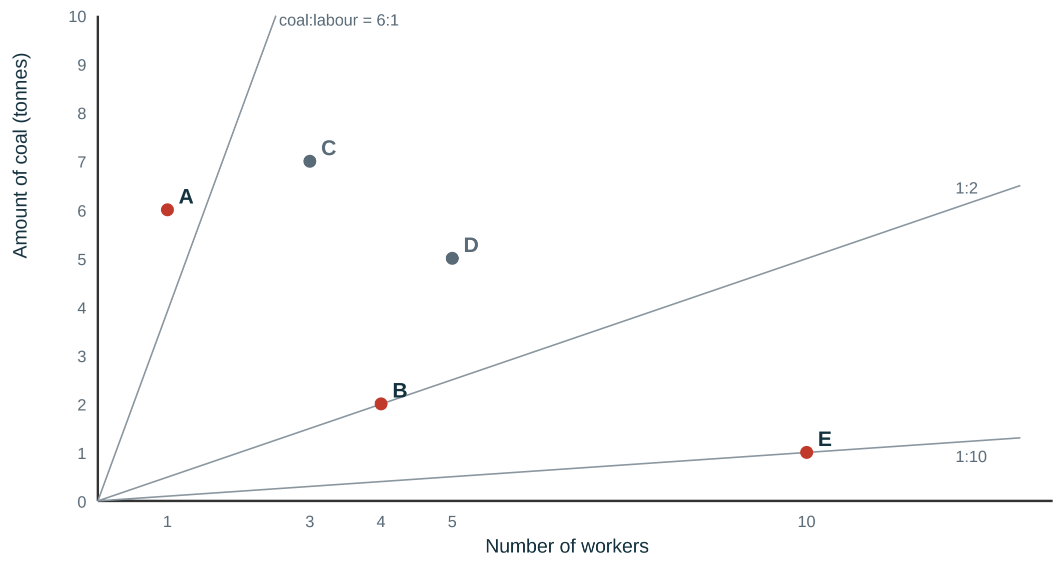

Figure 6. Five technologies (each produces the same 100 m of cloth). Rays from the origin show constant coal-to-labour ratios. Red points (A, B, E) will turn out to be the **sensible** choices; grey points (C, D) are wasteful — the next figure shows why.

Ruling out inferior (dominated) technologies

> **DEFINITION — DOMINATION**
>
> Technology 1 is **dominated** by technology 2 if technology 1 uses **at least as much of every input** (and more of at least one). A dominated technology will **never** be chosen, whatever the input prices are.

Compare the points: **C (3 workers, 7 coal)** needs **both** more workers *and* more coal than **A (1 worker, 6 coal) → C is dominated by A.** Likewise **D is dominated by B.** That leaves **A, B and E** — none of which is dominated (each is best on at least one input). To choose among *these*, we finally need **prices**.

Figure 7. Any technology lying **up-and-to-the-right** of A uses more of both inputs and is dominated by A. C sits in that region → eliminated. A, B and E survive.

CHECK YOURSELF

Is technology E dominated by anything? Why can a firm rationally pick the coal-heavy A *or* the labour-heavy E depending on circumstances? (Hint: nothing sits below-and-left of either.)

SECTION D · CONT.

Cost Minimisation & Isocost Lines

Firms want to maximise profit, which — for a given amount of cloth — means producing it at **the least possible cost**. The total cost of a technology using L workers and R tonnes of coal is:

$$\textit{cost} = (\textit{wage} \times \textit{workers}) + (\textit{coal price} \times \textit{tonnes}) = (w \times L) + (p \times R)$$

This is why the choice of technology depends on **economic information** — the **relative prices** of the inputs — and not on engineering alone.

Isocost lines

DEFINITION — ISOCOST LINE

An **isocost line** joins all input combinations that cost the **same total amount**. Rearranging the cost equation:

$$R = \frac{c}{p} - \frac{w}{p} \cdot L$$

Its **slope is −w/p**, the **relative price of the inputs**. Points **above** a line cost more; points **below** cost less. **Lower isocost lines = cheaper.** The firm picks the technology sitting on the **lowest attainable isocost line**.

Numerical anchor (wage £10, coal £20/tonne, so slope = −10/20 = −½):

Point	Workers	Coal (tonnes)	Total cost (£)
P_1	2	3	80
P_2	6	1	80
Q_1	3	6	150
Q_2	5	5	150

P_1 and P_2 lie on the **same £80 line**; Q_1 and Q_2 on the higher **£150 line**. Any point on a higher (further-out) line is more expensive.

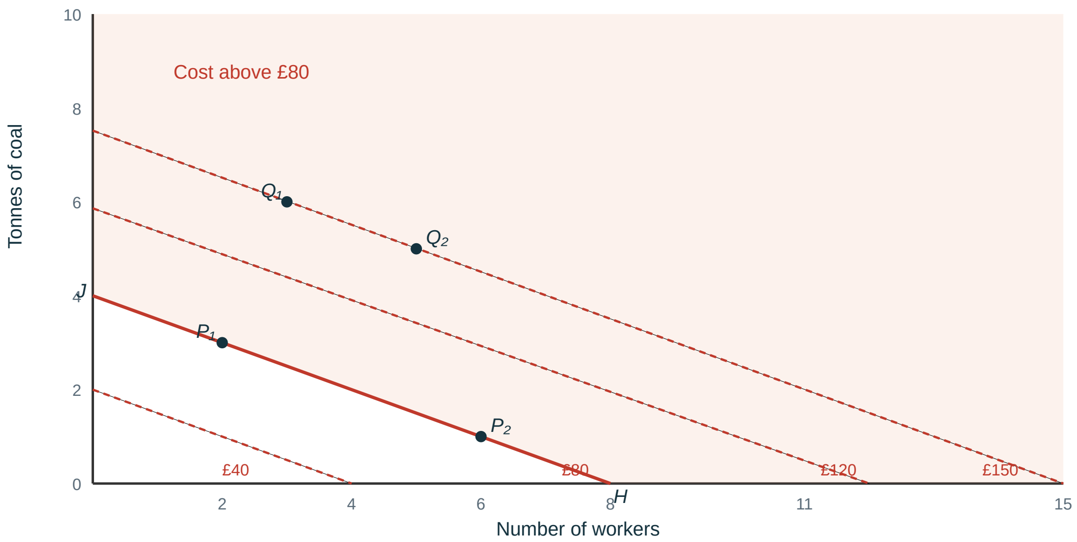

Figure 8. A family of parallel isocost lines (slope −w/p = −½). P_1 and P_2 share the £80 line; Q_1, Q_2 the £150 line. The firm wants the **lowest** line that still reaches a feasible technology.

Why Britain? Changing relative prices

Now put technologies and isocost lines **on the same graph**. Recall the surviving technologies for 100 m of cloth: **A = (1 worker, 6 coal)** — energy-intensive — and **B = (4 workers, 2 coal)** — labour-intensive.

Before the Industrial Revolution (coal £20, labour £10)

- Cost with **B**: (10×4) + (20×2) = **£80** → B sits on the lowest isocost line.
- Cost with **A**: (10×1) + (20×6) = **£130** → A is far more expensive.
- **Firms choose B.** There is **no incentive** to develop the coal-hungry technology A.

After coal gets cheaper (coal £5, labour £10)

The isocost slope becomes **−10/5 = −2** — the lines get **steeper**. Now:

- Cost with **A**: (10×1) + (5×6) = **£40**.
- Cost with **B**: (10×4) + (5×2) = **£50**.
- **A is now on the lowest isocost line — firms switch to the energy-intensive technology.**

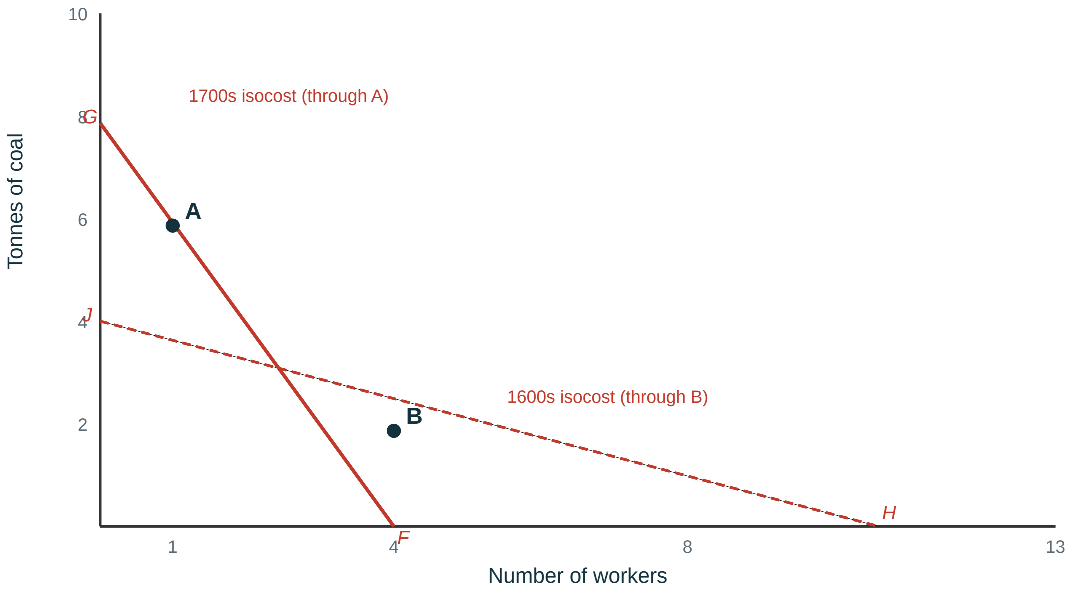

Figure 9. The **same two technologies**, two eras. In the **1600s** the flat isocost (cheap-ish labour relative to coal) makes **B** cheapest — A lies *outside* line HJ, so no one develops it. By the **1700s** labour is dear and coal cheap: the steep isocost makes **A** cheapest (B now lies outside FG). Britain's price signal is what triggered the switch.

THE BENEFIT OF INNOVATING — INNOVATION RENT

A firm that switches first to the newly-cheapest technology gains an advantage over rivals. Since `profit = revenue − costs`, the extra profit equals the **fall in cost**:

- Cost with old technology B: (4×10) + (2×5) = **£50**
- Cost with new technology A: (1×10) + (6×5) = **£40**

Innovation rent = £10 per 100 m of cloth. This rent is the incentive that pulls firms toward better technology.

Why Britain, and why then?

The model's answer is now sharp: the Industrial Revolution needed **(i) the capacity to innovate** and **(ii) relative prices that made energy-intensive technology cheapest**. Britain had both. Uniquely, **English wages were high** (relative to the cost of capital/energy) **and coal was cheap** — precisely the configuration that rewards labour-saving, coal-using machines.

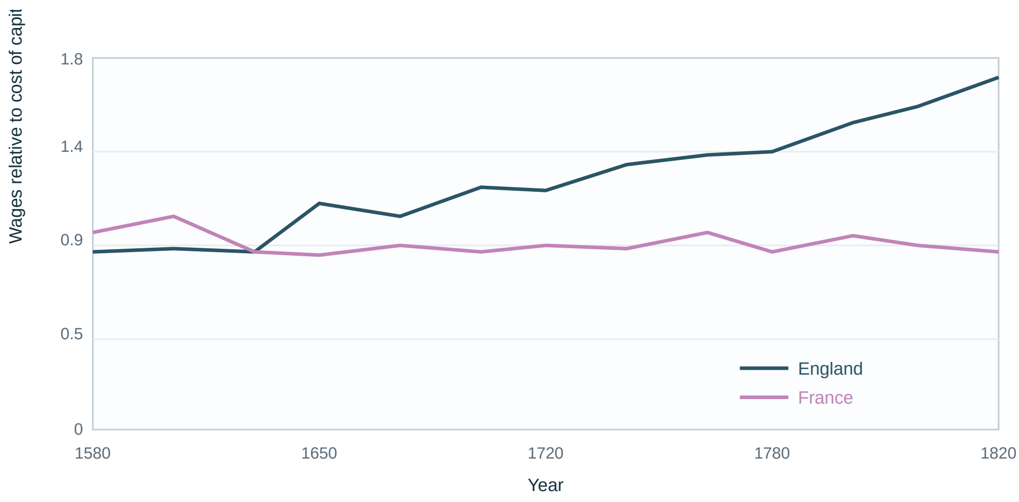

Figure 10. Wages relative to the cost of capital, 1580–1820. England's ratio climbs steadily (labour became expensive relative to machines/energy); France's stays flat and low. Only in England did the price signal steeply reward capital- and energy-intensive innovation — the answer to "why Britain first?"

Why did other countries eventually adopt it too?

- **Technological progress:** continued innovation reduced the inputs needed per unit (the technology moved from A toward an even leaner **A′**), lowering costs for everyone.
- **Falling energy costs and rising wages** spread internationally, so the coal-intensive technology eventually became cheapest elsewhere as well.

Entrepreneurs and creative destruction

DEFINITIONS — SCHUMPETER'S ENGINE

- **Entrepreneur** — the **first adopter**, the firm willing to try new technologies and start new businesses.
- **Schumpeterian (innovation) rents** — the temporary extra profits the first adopters enjoy.
- **Creative destruction** — the process by which old technologies and the firms that fail to adapt are **swept away** by the new, because they can no longer compete in the market.

The reward (innovation rent) and the threat (creative destruction) together make capitalism a **permanent innovation machine** — the mechanism absent from all the stagnant economies before 1800.

CHECK YOURSELF · EXAM MCQ

Two technologies A and B produce a car, with **machines on the Y-axis** and **workers on the X-axis**. Which statement is correct?

- (a) "A **dominates** B if, at current prices, a car is cheaper with A." — **False.** Domination means using *less of every input*; it is independent of prices. What (a) describes is A being *cheaper*, not dominating.
- (b) "If A is on an isocost line **above** B's, A is cheaper." — **False.** Higher (further-out) isocost lines cost *more*, so A would be dearer.
- (c) "Isocost lines become **flatter** if the price of machines goes up." — **True — this is the answer.** With machines on Y, the cost line is $M = c/p_m - (w/p_m)\cdot L$, so slope $= -w/p_m$. A higher machine price p_m shrinks the slope's magnitude → the lines get **flatter**.
- (d) "None of the above." — **False**, because (c) is correct.

Take-away: keep three ideas separate — **domination** (input quantities, price-free), **cost** (which isocost line you sit on), and **slope** (the input *price ratio*, read off the axis convention).

SECTION E

Economic Models

Everything we did in Sections C and D was **model-building**. This section steps back to ask: what *is* an economic model, why do we deliberately leave things out, and what makes one good?

Why do we need models?

An economy is the actions and interactions of **millions** of people. No one can hold all of that in mind at once. A **model** is a deliberate simplification that lets us see the **big picture**. Building one means separating:

- the **essential features relevant to the question** — kept in the model, and
- **unimportant details** — safely ignored.

KEY MINDSET

A model **omitting** detail is a **feature, not a bug.** The omissions are what make the important elements visible.

The metro-map analogy. A metro map is "wrong" about almost everything — distances, street layout, the curve of the lines. But for the one question a passenger actually asks — *"which line do I take, and where do I change?"* — it is far more useful than a geographically accurate map. A good economic model is exactly like this: it distorts the irrelevant to make the relevant unmistakable. (The Barcelona and Brussels metro maps on the slides make the same point — both are schematic, both are useful.)

Building a model — five steps

1. **Begin with a well-defined question.**
2. **Capture the elements** of the economy that matter for that question.
3. **Describe how agents act** and interact with each other and with the model's elements.
4. **Determine the outcomes** of these actions — an **equilibrium**.
5. **Study what happens when conditions change** (comparative statics).

Key ideas inside a model

VOCABULARY YOU MUST KNOW

- **Equilibrium** — a **self-perpetuating** situation: nothing of interest changes unless an **external force** alters the model's set-up.
- **Endogenous variable** — its value is **determined by the model**.
- **Exogenous variable** — its value is **determined outside the model** (taken as given).
- **Ceteris paribus** — "all else equal": study the relation between two variables while **holding everything else constant**.
- **Mathematics** — economic models are usually expressed with **equations and graphs**.

WORKED DISTINCTION — THE MALTHUS MODEL

In the Malthusian model, **population and living standards are endogenous** (the model determines them), while **technological progress is exogenous** (it arrives from outside). The whole story of the Industrial Revolution is what happens when that exogenous variable stops being a one-off shock and becomes **permanent**.

What makes a good model?

Four tests — a good model is:

- **Clear** — it helps us understand something important.
- **Predicts accurately** — its predictions are consistent with the evidence.
- **Improves communication** — it makes explicit what we agree and disagree about.
- **Useful** — we can use it to find ways to improve how the economy works.

CHECK YOURSELF

In the firm's technology-choice model of Section D, which variables are **exogenous** (given from outside) and which are **endogenous** (chosen by the firm)? *(Hint: input prices and the available technologies are exogenous; the technology actually chosen and the cost are endogenous.)*

SECTION F

Economic Growth & Climate Change

The same forces that produced the hockey-stick of prosperity also produced a hockey-stick of **carbon emissions**. This closing section connects the model to the defining challenge of the century.

Capitalism + carbon = growth + climate change

The Industrial Revolution was, at its core, an **energy transition**:

> **THE CENTRAL TRANSITION**
>
> From an economy where **photosynthesis** was the main energy source — so that **land was the binding constraint** on growth (the Malthusian world) — to an **energy-rich economy based on fossil fuels**, where that constraint was lifted.

- It produced **unprecedented increases in per-capita income**.
- It also produced **unprecedented increases in the Earth's surface temperature**.
- The ongoing reduction in global poverty **cannot** be achieved by the same "**coal-plus-capitalism**" recipe — the emissions would be catastrophic.

An unsustainable path

Across countries, richer means dirtier: CO_2 emissions per person rise strongly with GDP per person. Continuing to lift billions out of poverty along this line would push emissions far beyond safe limits.

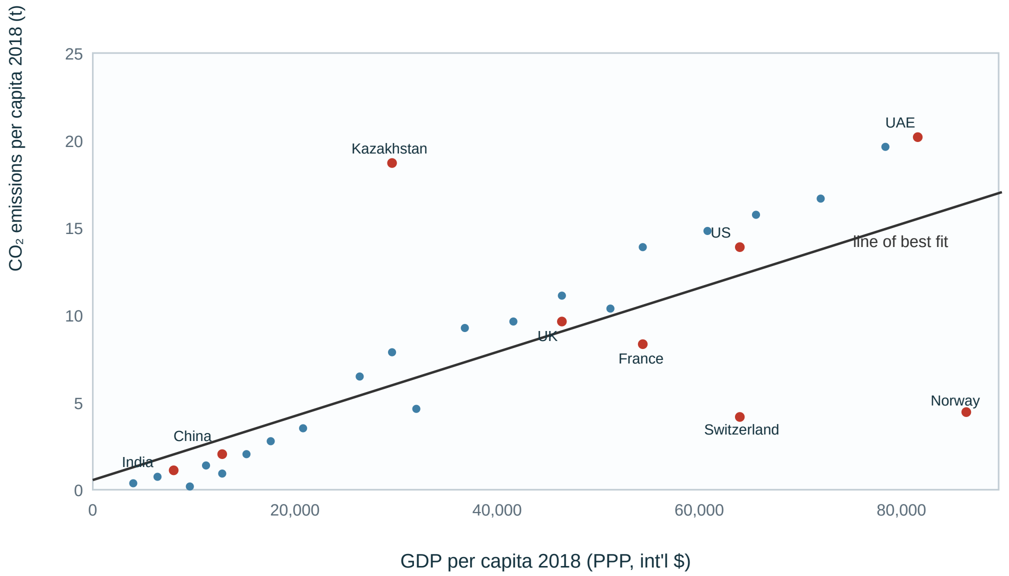

Figure 11. Richer countries emit more CO_2 per person (positive best-fit line). But the **scatter around the line matters**: France, Switzerland, Norway and Sweden are rich yet **low-carbon**, proving prosperity and low emissions *can* coexist — a matter of technology and energy mix, not income alone.

Some hope? Cheap clean energy and decoupling

The same innovation logic from Section D now works **for** the climate. As cumulative solar capacity grew, the price of solar modules collapsed along a steep **experience curve** — every doubling of installed capacity cut costs by a roughly constant percentage. Renewables have gone from luxury to **cheapest new electricity**.

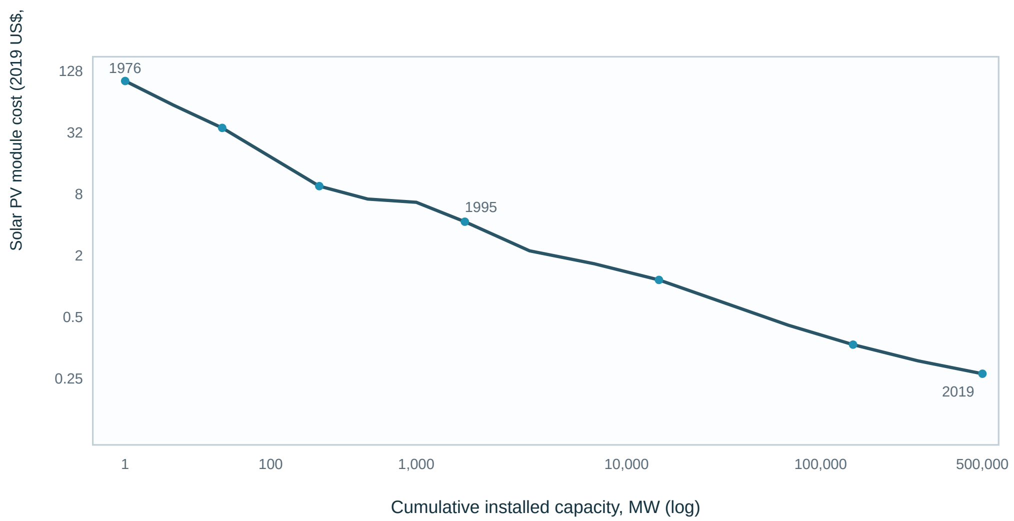

Figure 12. The solar-PV **experience curve** (log–log). Costs fell from ~$100/W (1976) to well under $0.50/W (2019) — a >99% decline — driven by cumulative deployment. This is **creative destruction working for decarbonisation**: the same innovation-rent mechanism, now pointed at clean energy.

There is also early evidence of **decoupling**: in several advanced economies, GDP per capita keeps rising while energy use per capita **falls** — even after adjusting for imported ("trade-adjusted") energy, so it is not merely offshoring emissions.

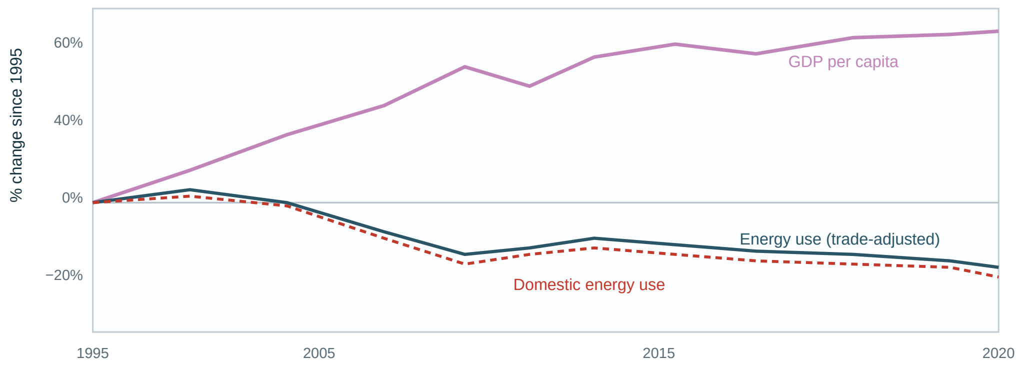

Figure 13. Relative decoupling: GDP per capita up ~54% since 1995 while energy use per capita is **down ~10%**. Because the fall holds even on a **trade-adjusted (consumption-based)** measure, it is genuine efficiency — not just outsourcing industry abroad.

LISTEN FOR IN THE LECTURE

The optimistic and pessimistic readings sit side by side. Pessimistic: the historical growth engine *runs on carbon* (Fig. 11). Optimistic: the *same* innovation mechanism can now make clean energy cheapest (Fig. 12) and break the income–energy link (Fig. 13). The exam question is usually about the **mechanism**, not which mood you pick.

WRAP-UP

Summary, Formulas & Glossary

The unit in one paragraph

Rational agents weigh benefits against the **full economic cost** (direct + opportunity), and the surplus over the best alternative is **economic rent**. **Specialization** — driven by **comparative advantage** — lets everyone consume beyond their own production frontier through **trade**. Firms choose **technologies** to minimise cost given input prices, represented by **isocost lines**; a change in **relative prices** (dear labour, cheap coal) made energy-intensive technology cheapest in **Britain**, and the **innovation rent** plus **creative destruction** turned one-off invention into permanent progress. **Models** are the deliberate simplifications that let us see all this. The same engine produced the climate crisis — and may yet produce its solution.

Formula sheet

Concept	Formula
Economic cost	`direct cost + opportunity cost`
Economic rent	`benefit − direct cost − opportunity cost = net benefit − opportunity cost`
Opportunity cost of a good	`(units of other good given up) — the slope of the PPF`
Comparative advantage in good X	`lower opportunity cost of X than the other party`
PPF (England)	`Clothes = 3600 − (6/5)·Wine`
PPF (Portugal)	`Clothes = 4000 − (8/9)·Wine`
Firm's cost	`cost = (w × L) + (p × R)`
Isocost line	`R = c/p − (w/p)·L   (slope = −w/p)`
Innovation rent	`cost(old technology) − cost(new technology)`
Profit	`profit = revenue − costs`

Glossary of terms to memorise

Term	Meaning
Opportunity cost	Net benefit of the next-best forgone alternative.
Economic vs accounting profit	Economic profit subtracts opportunity costs too; can be negative when accounting profit is positive.
Economic / innovation rent	Surplus over the best alternative; innovation rent = gain from a new technology vs the old.
Division of labour	Splitting production into specialised tasks (Smith's pin factory).
Learning by doing / economies of scale	Two engines of rising productivity from specialization.
Absolute advantage	Producing a good with fewer inputs than another party.
Comparative advantage	Producing a good at lower *opportunity* cost than another party.
Production Possibilities Frontier	All maximal output combinations for given resources; slope = opportunity cost.
Terms of trade	The price at which goods are exchanged; lies between the two opportunity costs.
Production function	Relation between inputs and output, Y = f(·).
Fixed vs variable proportions	Inputs must scale together vs. can be substituted.
Constant returns to scale	Doubling all inputs doubles output.
Dominated technology	Uses at least as much of every input as another → never chosen.
Isocost line	Equal-cost input combinations; slope = −relative input price.
Entrepreneur / creative destruction	First adopter earning Schumpeterian rents; laggard firms are swept away.
Model / equilibrium	Deliberate simplification; a self-perpetuating outcome.
Endogenous / exogenous	Determined inside / outside the model.
Ceteris paribus	"All else equal."

Exam-style questions to practise

1. A café owner reports €80,000 accounting profit but could earn €50,000 elsewhere and €40,000 by renting out her premises. Compute her economic profit and advise her.
2. Country X needs 10 hrs/car and 5 hrs/laptop; Country Y needs 8 hrs/car and 2 hrs/laptop. Who has the comparative advantage in cars? Show the opportunity costs.

3. Two technologies produce steel: A = (2 workers, 8 coal), B = (6 workers, 3 coal). At wage 20 and coal 10, which is cheaper? At wage 20 and coal 30? Explain using isocost slopes.
4. Explain, with a diagram, why the Industrial Revolution's energy-intensive technologies were adopted in Britain before France.
5. Define an equilibrium and give the endogenous and exogenous variables of the Malthusian model.
6. "Prosperity requires high emissions." Use Figures 11–13 to argue both for and against.

Study guide prepared as a companion to *Microeconomics — Prosperity and Inequality, Unit 2: Technological Change, Population, and Growth* (KU Leuven, 2025–2026). All diagrams are clean redrawings of the lecture figures for study purposes. Cross-references such as "extension 2.4 in the handbook" point to the CORE / course handbook.